



UTM
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SECTION 1

PROJECT TITLE:

**IMPACT ANALYSIS OF GEOGRAPHIC, ECONOMIC, AND SEASONAL
DETERMINANTS ON RETAIL SALES PERFORMANCE**

PHASE 1

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1.0 Introduction

This phase focuses on developing interactive dashboards using Microsoft Power BI to analyze retail sales performance and identify factors influencing sales. The dashboard was designed to transform the processed data from Phase 2 into meaningful visualizations that support business decision-making.

The dashboard consists of two main pages:

1. **Executive Sales Overview**
2. **Sales Influencing Factors**

Interactive slicers such as Year, Store Type, Performance Tier, Temperature Band, and Holiday Status were included to allow users to dynamically explore the data and gain deeper insights.

2.0 Dashboard 1: Executive Sales Overview

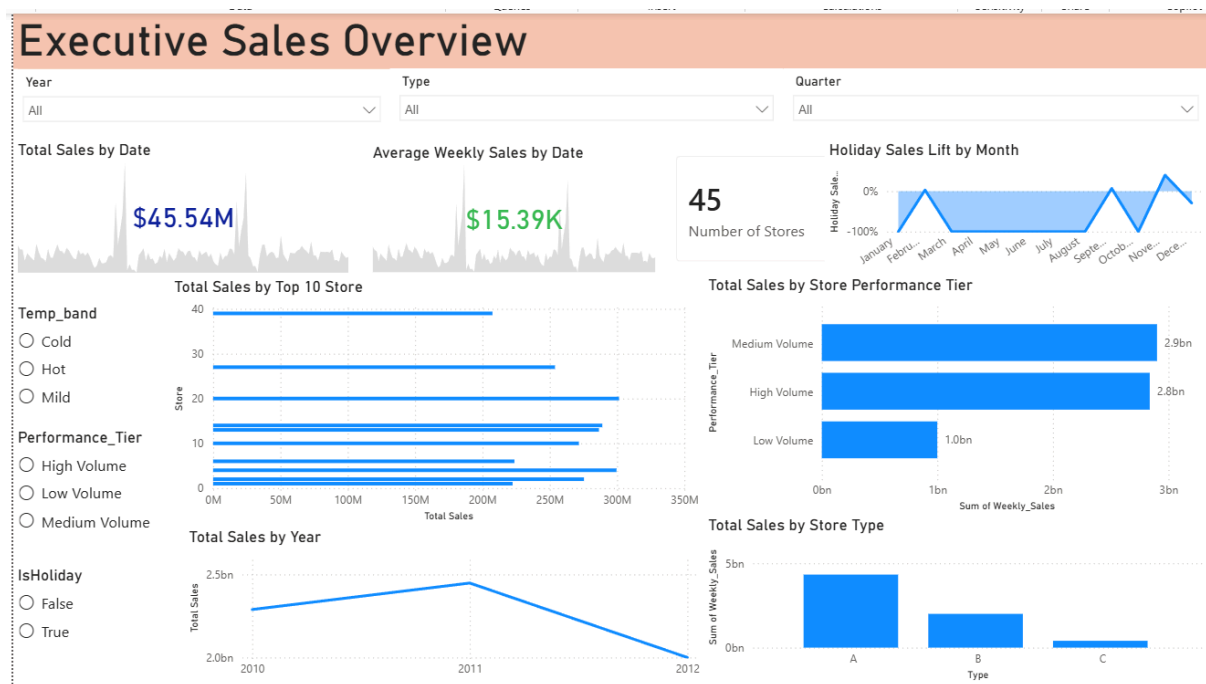


Figure 1: Executive Sales Overview Dashboard

Dashboard Overview

This dashboard provides an overall view of sales performance across stores and years. It focuses on key performance indicators (KPIs), sales trends, and comparisons among different store categories.

Key Visualizations

1. The KPI Cards

The KPI cards give a high-level summary of total sales, weekly averages, store count, and holiday lift. Through the KPI cards, an instant snapshot of the company health and baseline performance can be easily captured by executives before they dive into the deeper trend and location-specific data.

A. Total Sales KPI

The Total Sales KPI indicates that the retail chain generated approximately **\$45.54 million in total revenue** during the analyzed period, establishing the baseline scale of the business.

B. Average Weekly Sales KPI

The Average Weekly Sales KPI shows that stores achieved an average of approximately **\$15.39 thousand per week**, providing a benchmark for evaluating store performance.

C. Number of Stores KPI

The dataset contains **45 stores**, which forms the basis of the sales analysis.

D. Holiday Sales Lift By Month KPI

This KPI breaks down the percentage difference in sales between holiday and non-holiday periods across a monthly timeline. It highlights exactly which months experience a massive revenue surge due to holiday events and which months remain flat, allowing management to see the direct seasonal impact on monthly performance.

2. Total Sales by Top 10 Store

This chart filters and compares sales performance among the top ten individual locations. Several stores consistently outperform the rest of the fleet, which might be indicating a strong local customer demand, store location effectiveness, or larger store capacity.

3. Total Sales by Store Performance Tier

Stores were categorized into High Volume, Medium Volume, and Low Volume tiers based on sales performance.

Key findings:

- Medium Volume stores generated the highest overall sales.
- High Volume stores contributed a similar amount of revenue.
- Low Volume stores generated significantly lower sales.

Total Sales by Store Performance Tier graph indicates that revenue is concentrated among medium and high-performing stores. The graph shows medium as the highest shows that most stores have medium sales volume.

4. Total Sales by Store Type

Stores are categorised by size, A being the large stores, B the medium sized stores and C is the small stores. Referring to Total Sales by Store Tier, Store Type A generated the highest revenue, followed by Type B and Type C.

This suggests that larger store formats contribute more significantly to total company revenue.

5. Total Sales by Year

Sales increased from 2010 to 2011 before experiencing a decline in 2012. This macro trend over the three-year period may indicate changing economic conditions, shifting market demand, or a shorter data collection window in the final year.

Dashboard 1 Insights

- Total sales exceeded \$45 million during the study period.
- Type A stores are the strongest contributors to revenue.
- Medium and High Volume stores account for most sales.
- Sales peaked in 2011 before declining in 2012.
- Store performance varies significantly across locations.

3.0 Dashboard 2: Sales Influencing Factors

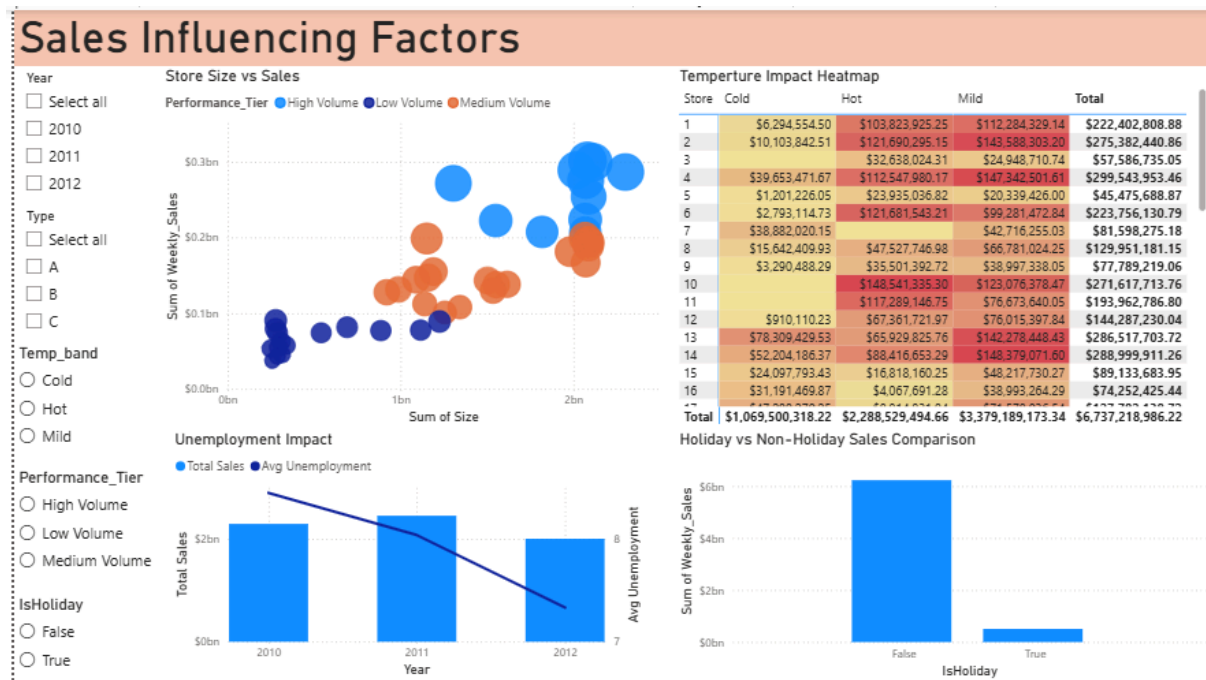


Figure 2: Sales Influencing Factors Dashboard

Dashboard Overview

This dashboard investigates the factors that may influence retail sales, including store size, holidays, temperature conditions, and unemployment levels.

Key Visualizations

1. Store Size vs Sales Scatter Plot

The scatter plot shows a positive relationship between store size and sales performance.

Key findings:

- Larger stores generally achieve higher sales.
- High Volume stores are concentrated in the upper-right region of the chart.
- Low Volume stores are mostly associated with smaller store sizes and lower sales.

This suggests that store size is an important factor influencing sales performance.

2. Holiday vs Non-Holiday Sales Comparison

The chart compares sales generated during holiday and non-holiday periods.

Key findings:

- Non-holiday periods account for the majority of total sales.

- Holiday sales represent a smaller portion of overall revenue due to fewer holiday weeks in the dataset.

We can see that non-holiday sales are higher, this is because most days are not holiday compared to the one that is. However, holiday periods still produce noticeable sales spikes and remain important for promotional planning.

3. Temperature Impact Heatmap

The heatmap compares total store sales performance under different temperature conditions which consists of cold, mild, and hot.

Key findings:

- Most of the stores experienced higher sales activities in mild and hot conditions compared to cold conditions.
- Some of the stores demonstrate a high concentration of sales in the hot and mild temperature categories, showing that these environments are more conducive to the specific consumer demands that are managed by its locations.
- The heatmap highlights that sales revenue is not evenly distributed across all temperature bands, indicating that the external weather conditions may influence shopping behavior and seasonal demand.

4. Unemployment Impact

It uses line and clustered column charts to correlate the average of unemployment with total sales performance over time.

Key findings:

- The average unemployment rate shows a consistent decrease from approximately 8.45 in 2010 to 7.33 in 2012.
- While the unemployment rate consistently declines, the sales performance increases to approximately 2.5 billion dollars in 2011 before dropping in 2012, suggesting that while the macro-economic trend is positive, the sales drivers may be influenced by other external factors.
- This pattern shows that the economic conditions likely influence customer spending behaviours.

Dashboard 2 Insights

- Store size positively influences sales performance.
- Holidays affect sales patterns and create opportunities for promotions.
- Weather conditions may contribute to variations in customer demand.
- Economic factors such as unemployment can impact purchasing behavior.

4.0 Storyboard Summary

Story 1: Store Size Drives Sales

The analysis shows a strong relationship between store size and sales performance. Larger stores tend to generate higher revenue due to greater product variety and customer capacity.

Story 2: Revenue Is Concentrated in Stronger Stores

Most sales are generated by Medium Volume and High Volume stores. Low Volume stores contribute a smaller portion of total revenue.

Story 3: Store Format Matters

Type A stores consistently outperform Types B and C, suggesting that larger store formats offer better sales opportunities.

Story 4: External Factors Influence Sales

Temperature, holidays, and unemployment levels all show varying degrees of influence on sales performance. Businesses should monitor these factors when planning operations and marketing campaigns.

5.0 Business Recommendations

Based on the dashboard findings, the following recommendations are proposed:

1. Prioritize High-Performing Stores

Management should allocate more resources and inventory to Medium Volume and High Volume stores, as these stores contribute the largest share of revenue.

2. Expand Successful Store Formats

Since Type A stores generate the highest sales, future expansion strategies should prioritize this store format where feasible to maximize returns of investment.

3. Leverage Holiday Promotions

Targeted promotional campaigns should be implemented during holiday periods to maximize customer spending and increase revenue.

4. Optimize Store Layout and Capacity

The positive relationship between store size and sales suggests that increasing selling space and improving store layouts may enhance the store overall performance.

5. Monitor Economic and Environmental Factors

Management should continuously monitor unemployment trends and seasonal weather conditions to adjust pricing, inventory, and marketing strategies accordingly.

6.0 Conclusion

Overall, the Power BI dashboards have successfully transformed the retail sales data into actionable business insights through interactive visualizations and KPIs. The analysis revealed that sales performance is significantly impacted by a combination of operational factors such as store size, store type, performance tier, also the external factors like holidays, and economic conditions. These findings can support management in making data-driven decisions regarding resource allocation, expansion planning, and marketing strategies to improve overall business performance.